ALCHEMUS

Advanced Drug Analysis Platform

Research Owner Wallet

6cTDF4hLRDTcPodcf5uAAxDrYf51DSpJV92LiDorj2Cu

Date: 5/1/2025

Compounds: Psilocybin, Laughing Gas, Flunitrazepam, Salbutamol

New Compound Analysis

Name: Fluniprosal

Formula: $C_{20}H_{20}FN_3O_5$

Weight: 397.39

Drug-likeness: 0.85

Activity:

sedative, anesthetic, psychedelic, bronchodilator

Target Analysis

GABA-A receptor

Binding Affinity: -9.5

Widely implicated in anxiety, sleep disorders, and sedation application.

5-HT2A receptor

Binding Affinity: -8

Important in mood regulation and cognition, connected to the effects of hallucinogens.

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Binding Affinity: -9

Involved in bronchodilation, a key therapeutic target in asthma management.

Toxicity Profile

Risk Level: moderate

Affected Systems: central nervous system, respiratory system

Mitigation Suggestions:

- monitoring dosage
- regular CNS function assessment
- screening for psychological disorders before use
- combining with cognitive therapies

Pharmacokinetics

Absorption:
rapid gastrointestinal absorption

Distribution:
wide distribution with high lipophilicity

Metabolism:
hepatic via CYP enzymes

Excretion:
renal excretion of metabolites

Half-life: approximately 12 hours

Detailed Analysis

Fluniprosal is an innovatively designed compound that amalgamates the psychotropic effects of psilocybin with the sedative properties of flunitrazepam, alongside the bronchodilatory action of Salbutamol. This hybrid molecule is tailored to simultaneously address a range of conditions. Its pharmacodynamic profile is anticipated as a result of synergizing multiple receptor modulations. Critical assessment of drug-likeness indicates widespread applicability with moderate associated risks, necessitating a strategic approach for clinical approval, particularly in the context of safety and efficacy. The framework for research indicates a robust plan for interdisciplinary exploration from molecular composition to application, aiming for significant therapeutic breakthroughs in CNS and respiratory medication landscapes.
